

Chapter 2 - Classification of finite reflection groups

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September 17, 2026

2 Irreducible components

Exercise 2.1 [Skipped]

Let W be the dihedral group $\mathcal{D}_6$ of order 12, with standard Coxeter generators $S = \{s, s'\}$. The Coxeter system (W, S) is irreducible. However, W has another set S' of Coxeter generators leading to a Coxeter system which is not irreducible: $S' := \{s, (s's)^3, s(s's)^2\}$.

Remark 2.1

This tells us that irreducibility is a property of the Coxeter system, i.e. of the chosen set S , not of the abstract group W .

8 Crystallographic groups

Exercise 8.1

If W satisfies the necessary condition in the proposition (and is essential), show how to modify the lengths of the roots in a simple system Δ so that the resulting $\mathbf{Z}$ -span is a lattice in V stable under W . [Use the fact that the Coxeter graph has no circuits.]

Solution. For any two $\alpha_i, \alpha_j \in \Delta$, we'll modify their length so that $2 \frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} \in \mathbb{Z}$, in which case

$$s_{\alpha_i}(\alpha_j) = \alpha_j - 2 \frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} \alpha_i$$

lies in the $\mathbb{Z}$ -span of α_i, α_j .

Notice

$$2 \frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} = \frac{|\alpha_j|}{|\alpha_i|} \cdot 2 \cos \langle \alpha_i, \alpha_j \rangle$$

while

$$2 \frac{(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)} = \frac{|\alpha_i|}{|\alpha_j|} \cdot 2 \cos \langle \alpha_i, \alpha_j \rangle$$

And since $m(\alpha, \beta) \in \{2, 3, 4, 6\}$ since W is crystallographic,

$$\cos \langle \alpha_i, \alpha_j \rangle = \cos \frac{\pi}{m(\alpha_i, \alpha_j)} \in \left\{ 0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2} \right\}. \text{ (0 is omitted as } \alpha_i \neq \alpha_j \text{)}$$

We can have $2 \cos \langle \alpha_i, \alpha_j \rangle = \sqrt{n_{ij}}, n_{ij} \in \mathbb{Z}$. Now, we may let $\frac{|a_i|}{|a_j|} = \sqrt{n_{ij}}$, and therefore

$$2 \frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)} = \frac{1}{\sqrt{n_{ij}}} \sqrt{n_{ij}} = 1 \in \mathbb{Z}$$

$$2 \frac{(\alpha_j, \alpha_i)}{(\alpha_j, \alpha_j)} = \sqrt{n_{ij}} \sqrt{n_{ij}} = n_{ij} \in \mathbb{Z}.$$

(Following from the hint)

Now, we may pick out any $\alpha_i \in \Delta$ as the starting vertex, unify it to length 1, then scale all the vertices, denote one of them by α_j , in its neighbourhood to the length satisfying $\frac{|\alpha_i|}{|\alpha_j|} = \sqrt{n_{ij}}$. Since there's no loop in the corresponding Coxeter Graph, such process can go through all the vertices in this finite Coxeter Graph without contradiction, and the $\mathbb{Z}$ -span was obtained.